



EmotionCam User Manual

Local-first visible-expression estimates with optional external AI analysis

Version 1.1.0 AI-enabled | Updated June 16, 2026



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Local-first visible-expression estimates with optional external AI analysis

Updated June 16, 2026

EmotionCam estimates visible facial expressions. It does not know or guarantee a person's true emotions.

1. Introduction

EmotionCam is a Windows desktop app for one person using a laptop webcam. It shows a local camera preview, estimates visible facial expressions, smooths the result, and displays a label, group, confidence estimate, safe interaction message, and recent timeline.

EmotionCam can also be personalized through local calibration, run in the Windows tray, save metadata-only logs, show local Statistics, optionally send daily text summaries when explicitly enabled, and optionally use external AI analysis after explicit consent and API-key setup.

2. Privacy and Safety

- Webcam frames are processed locally by default.
- External AI analysis is disabled by default and cannot send images unless the user enables it, accepts the consent checkbox, and provides an OpenAI API key.
- In External AI mode, EmotionCam sends one selected cropped face image by

default. Full-frame sending is available only if the user changes the setting, and may include background details.

- EmotionCam does not identify people or perform face matching.
- There is no account system, telemetry, analytics, automatic screenshot, or video-recording feature.
- Metadata logs contain expression labels/groups, confidence, status, messages, popup status, FPS, AI status fields, and final result source.
- Calibration stores local normalized feature data and labels.
- Raw calibration frames are not saved unless the explicit debugging option is enabled.
- Daily email summaries are off by default and send text only if enabled.
- API keys are not stored in config.json. Secure storage uses Windows credential storage through keyring when available; otherwise the key can be used for the current session only.

Do not use EmotionCam for medical, psychological, legal, hiring, security, surveillance, or lie-detection decisions.

3. Installation

Install the packaged app

- 1 Open the repository Releases page.
- 1 Download EmotionCam_Setup.exe for v1.0.0 local-only or EmotionCam_Setup_v1.1.0_AI.exe for the AI-enabled release.
- 1 Run the installer.
- 1 Launch EmotionCam from the Start Menu or optional desktop shortcut.
- 1 Allow Windows camera access for desktop apps if prompted.

The app installs per user under:

%LOCALAPPDATA%\Programs\EmotionCam

The installer is unsigned, so Windows SmartScreen may show a warning. Choose More info > Run anyway only if you trust the downloaded file.

Which release to choose

Choose v1.0.0 Local-only if privacy/offline use matters most and you do not want the External AI option.

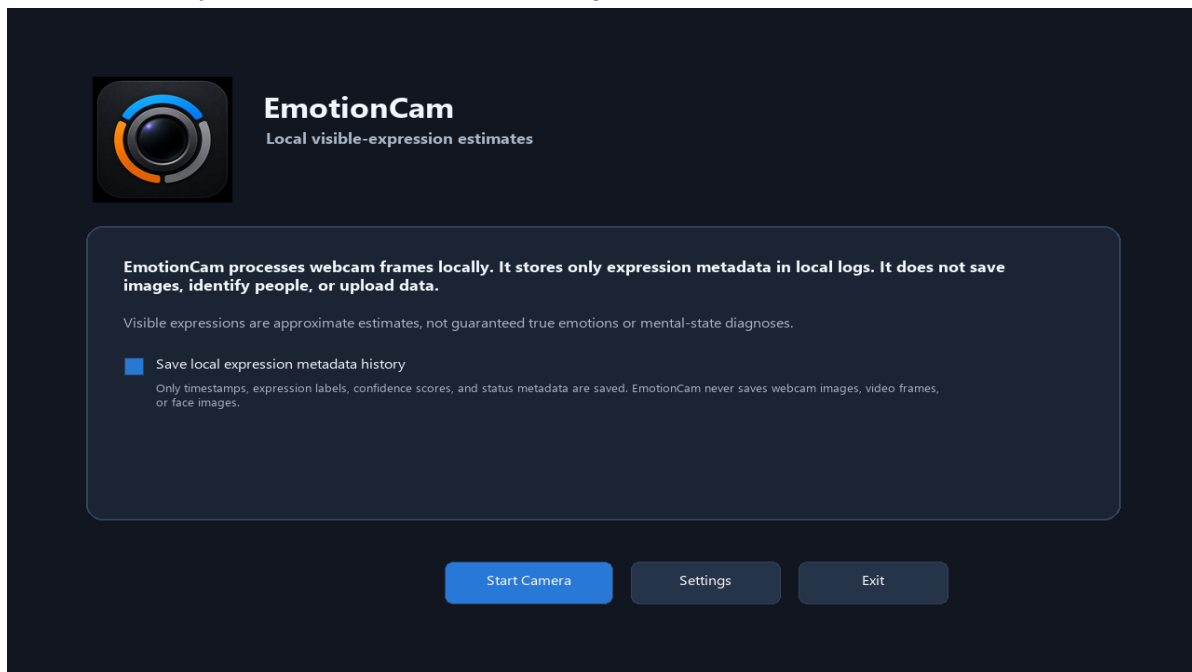
Choose v1.1.0-ai AI-enabled if you want all local features plus optional AI-assisted visible-expression analysis and accept that enabling External AI may send selected cropped face images or frames to OpenAI.

Run from source

```
py -3.12 -m venv .venv
.\.venv\Scripts\Activate.ps1
python -m pip install --upgrade pip
python -m pip install -r requirements.txt
python -m app.main
```

4. First Launch

The startup screen shows the local-processing privacy notice, approximation notice, metadata-history checkbox, Start Camera, Settings, and Exit.



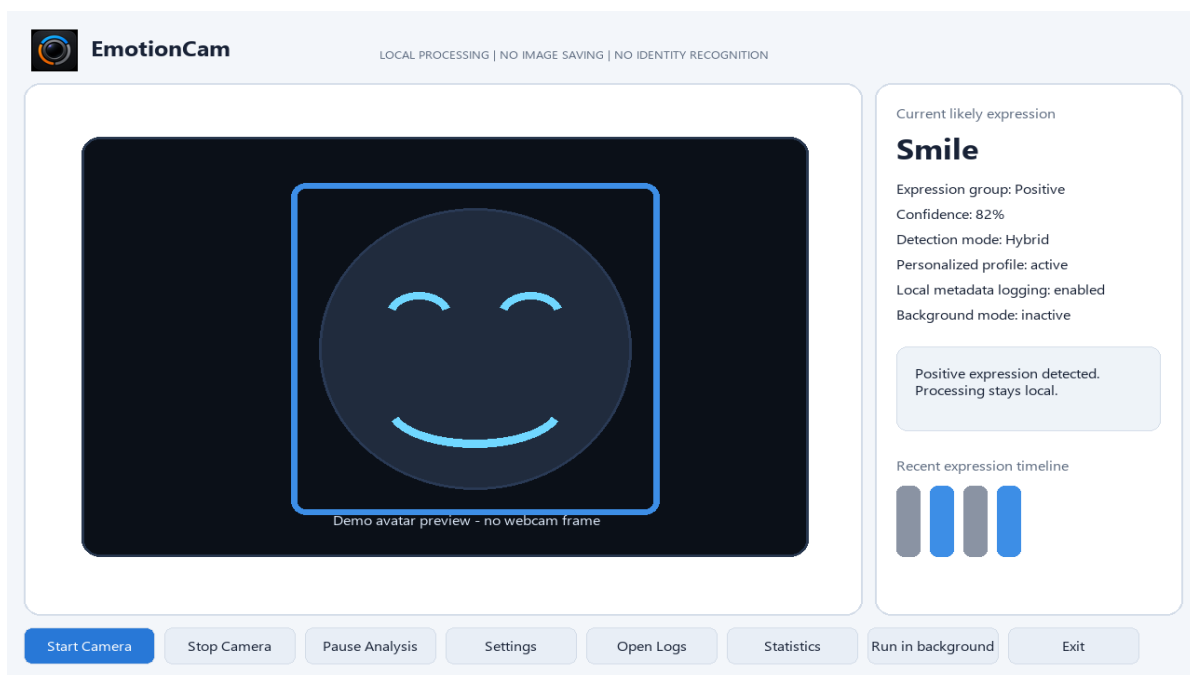
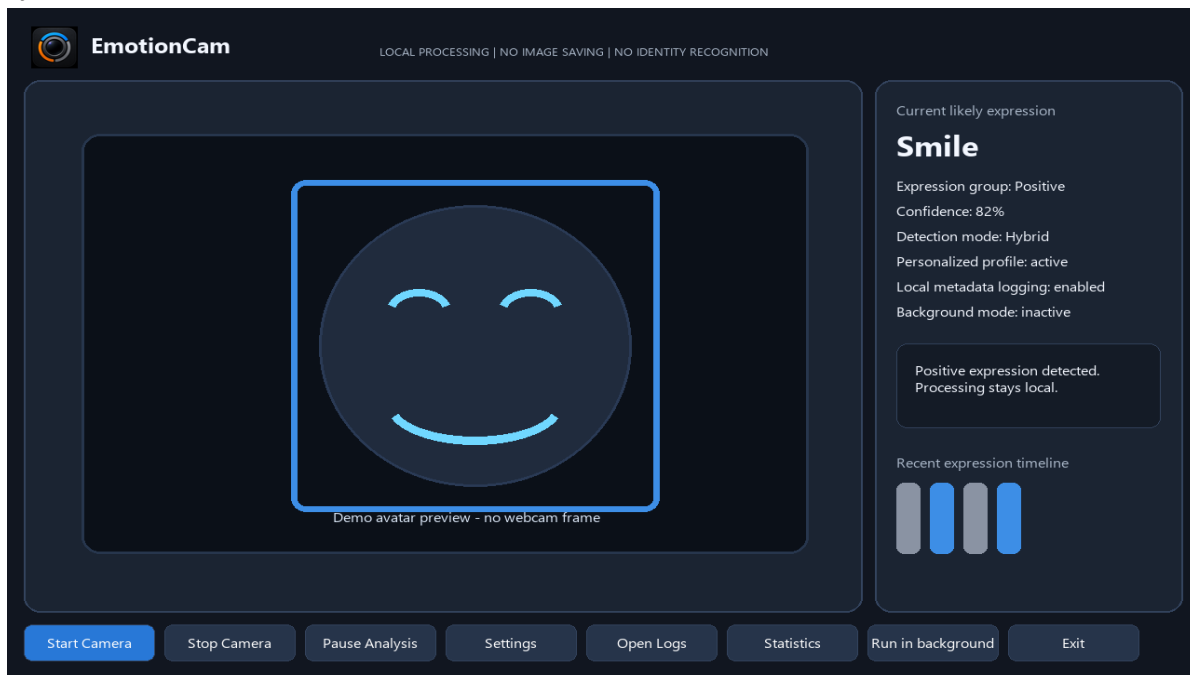
5. Main Dashboard

The dashboard contains the camera preview, face rectangle, current likely expression, expression group, confidence, interaction message, timeline, logging status, privacy status, detection mode, personalized profile status, External AI status, Statistics, Settings, Open Logs, Run in background, and Exit controls.

Rectangle colors use the smoothed expression group:

- Blue: positive

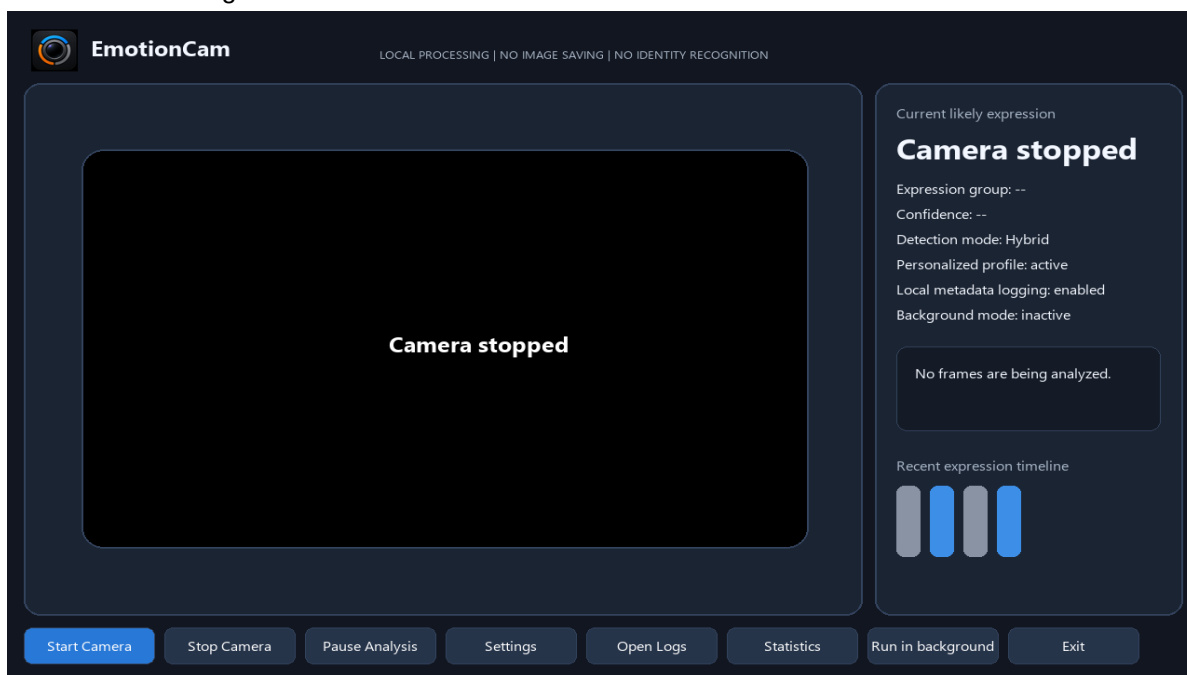
- Orange: negative
- Gray: neutral, unknown, low confidence, or no face



6. Starting and Stopping the Camera

- 1 Click Start Camera.
- 1 Wait for the dashboard preview and local analysis status.
- 1 Use Pause Analysis to pause classification while the preview remains active.

- 1 Click Stop Camera to stop analysis, release the camera, and clear the last frame.
- 1 The preview shows a black Camera stopped screen.
- 1 Click Start Camera again to restart.



7. Expression Detection

Heuristic labels include neutral, happy, smile, laughing, surprised, sad, angry, fearful, disgusted, confused, focused, tired, unknown, low_confidence, and no_face.

Personalized calibration also supports smile_small, smile_big, amused, bored, frustrated, concerned, skeptical, thinking, and relaxed.

Group	Labels
Positive	happy, smile, smile_small, smile_big, laughing, amused, surprised
Negative	sad, angry, fearful, disgusted, tired, bored, frustrated, concerned
Neutral	neutral, focused, confused, skeptical, thinking, relaxed, unknown, low_confidence, no_face

Confidence is an estimate. Temporal smoothing and stable-expression timing reduce flicker. Strong negative labels require higher confidence, while uncertain results fall back toward neutral or unknown.

Detection modes include Local only, Personalized only, Personalized / Hybrid local, External AI only, and Hybrid local + AI.

8. External AI Analysis

External AI Analysis is available in the AI-enabled release and uses the OpenAI vision API to estimate the visible expression in one selected image. It may help validate or improve local estimates, but it is still approximate and does not know true internal emotions.

External AI is off by default. To enable it:

- 1 Open Settings.
- 1 Go to Expression Detection Mode and External AI.
- 1 Choose External AI only or Hybrid local + AI.
- 1 Check Enable External AI Analysis.
- 1 Read and accept the consent checkbox.
- 1 Enter an OpenAI API key.
- 1 Keep Send cropped face only enabled unless you intentionally want to send the selected full frame.
- 1 Click Test Connection.
- 1 Save Settings.

Consent is required even if the checkbox is accidentally enabled. If consent is missing, no request is sent. If the API key is missing, invalid, unavailable, or the request times out, EmotionCam falls back to local detection.

The AI request sends only one selected image at a controlled interval. The default interval is 10 seconds, with a minimum of 5 seconds. It never sends every webcam frame. The request asks for structured JSON with a visible-expression label, group, confidence, short reason, and alternatives. EmotionCam validates the JSON, accepts only known labels/groups, clamps confidence to 0-1, and falls back to unknown when the result is invalid or too uncertain.

What may be sent when External AI is enabled:

- One cropped face image by default.
- One selected full frame only if cropped-face-only mode is turned off.

What is never sent or logged:

- API keys.
- Raw base64 image data in logs.
- Saved webcam video.

- Face identity data.
- Face matching data.
- Medical or psychological conclusions.

Settings > Expression Detection Mode and External AI

External AI analysis sends selected cropped face images or selected frames to an external AI service. It is disabled by default and requires explicit consent.

Detection mode	Local only
Enable External AI Analysis	Off
Consent accepted	No - no images are sent
API provider	OpenAI
OpenAI API key	***** (not shown)
Send cropped face only	On - recommended
AI request interval	10 seconds
AI timeout	20 seconds
Show AI debug info	Off

Test Connection

Settings > Expression Detection Mode and External AI

External AI analysis sends selected cropped face images or selected frames to an external AI service. It is disabled by default and requires explicit consent.

Detection mode	Hybrid local + AI
Enable External AI Analysis	On
Consent accepted	No - no images are sent
API provider	OpenAI
OpenAI API key	***** (not shown)
Send cropped face only	On - recommended
AI request interval	10 seconds
AI timeout	20 seconds
Show AI debug info	Off

Test Connection

Settings > Expression Detection Mode and External AI

External AI analysis sends selected cropped face images or selected frames to an external AI service. It is disabled by default and requires explicit consent.

Detection mode	Hybrid local + AI
Enable External AI Analysis	On
Consent accepted	Yes
API provider	OpenAI
OpenAI API key	 (not shown)
Send cropped face only	On - recommended
AI request interval	10 seconds
AI timeout	20 seconds
Show AI debug info	Off

Test Connection



EmotionCam

LOCAL BY DEFAULT | EXTERNAL AI ENABLED BY CONSENT | NO IMAGE SAVING



Demo avatar preview - no webcam frame

Current likely expression

Smile Small

Expression group: Positive

Confidence: 78%

Detection mode: External AI only

Personalized profile: active

External AI: enabled | Missing API key

AI result: --

Final source: local fallback

If no key is configured,
EmotionCam keeps using local
detection.

Start Camera

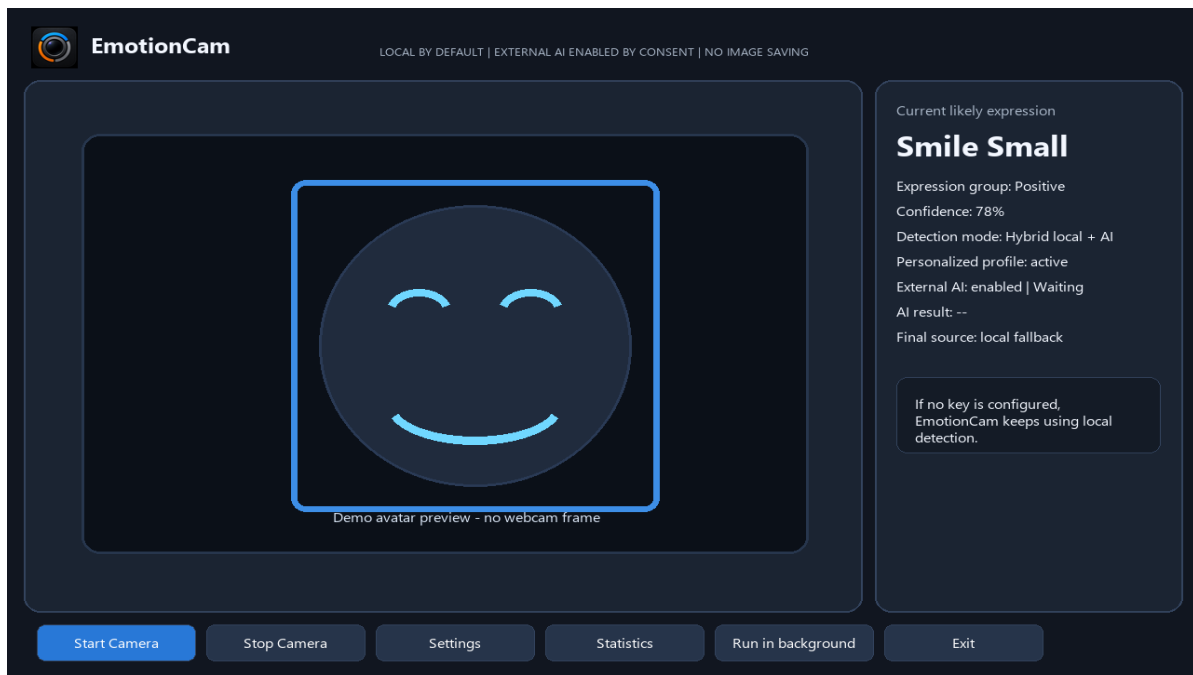
Stop Camera

Settings

Statistics

Run in background

Exit



9. Personalized Calibration

Open calibration from Improve detection with calibration or Settings >

Train / Calibrate Expressions.

The calibration screen includes:

- Target expression dropdown
- Local example expression graphic
- Demo/live preview area
- Live detected-expression estimate
- Match helper/checkmark
- Previous, Start Capture, Re-capture, Next, and Finish
- Progress, valid sample count, rejected sample count, and quality summary

The old Skip button and separate Label Unknown Expression flow were removed. Users now choose the target expression directly from the dropdown.

Personalized expression calibration

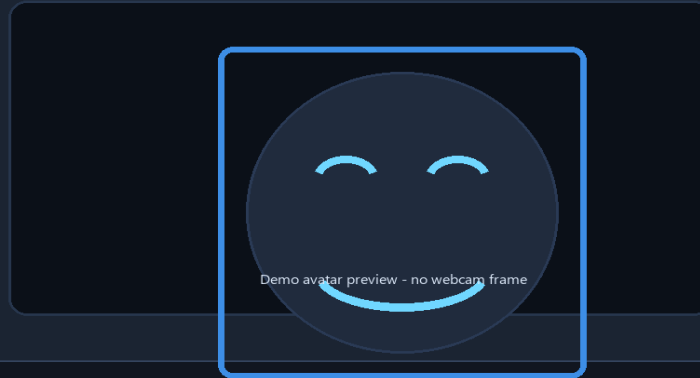
Calibration stores local expression feature data and labels. EmotionCam does not save camera images unless debug image storage is explicitly enabled.

Target expression Smile Small ▾

Example expression graphic



Match helper: Preparing



Valid samples: 32 | Rejected samples: 3 | Quality summary: good

Previous

Start Capture

Re-capture

Next

Finish

Personalized expression calibration

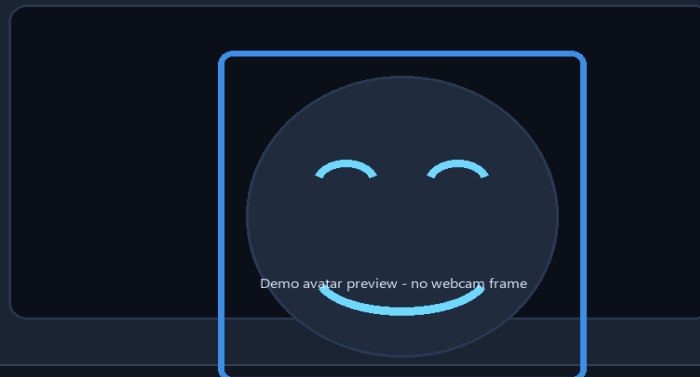
Calibration stores local expression feature data and labels. EmotionCam does not save camera images unless debug image storage is explicitly enabled.

Target expression Surprised ▾

Example expression graphic



Match helper: Preparing



Valid samples: 32 | Rejected samples: 3 | Quality summary: good

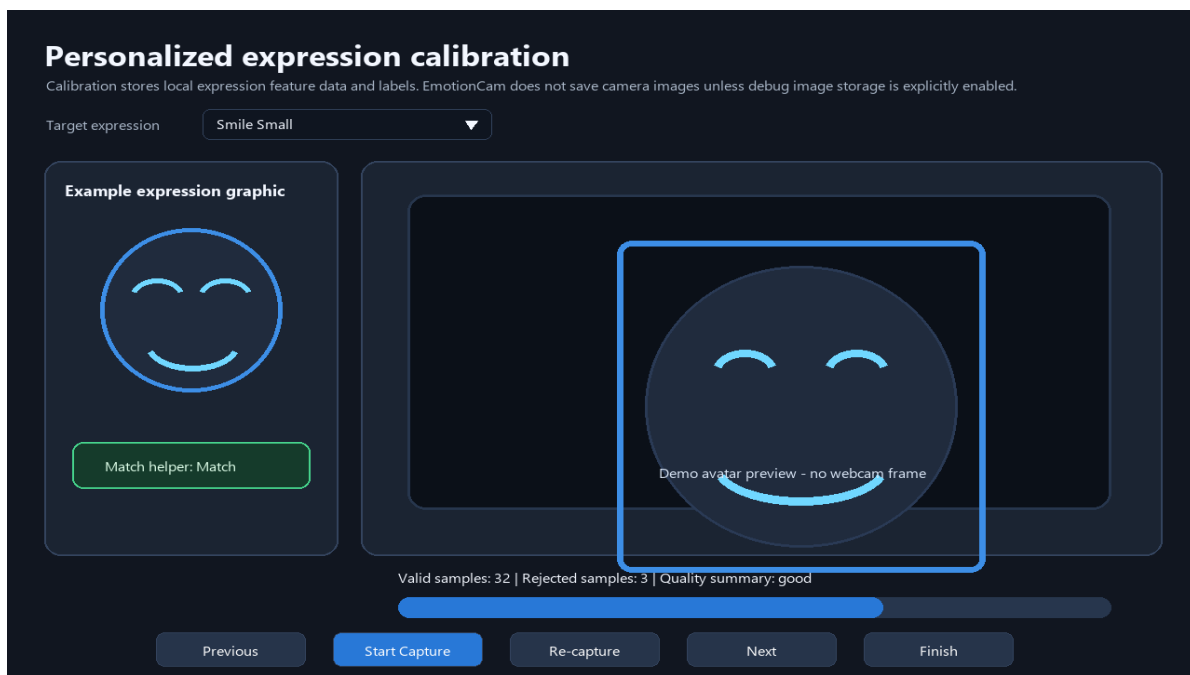
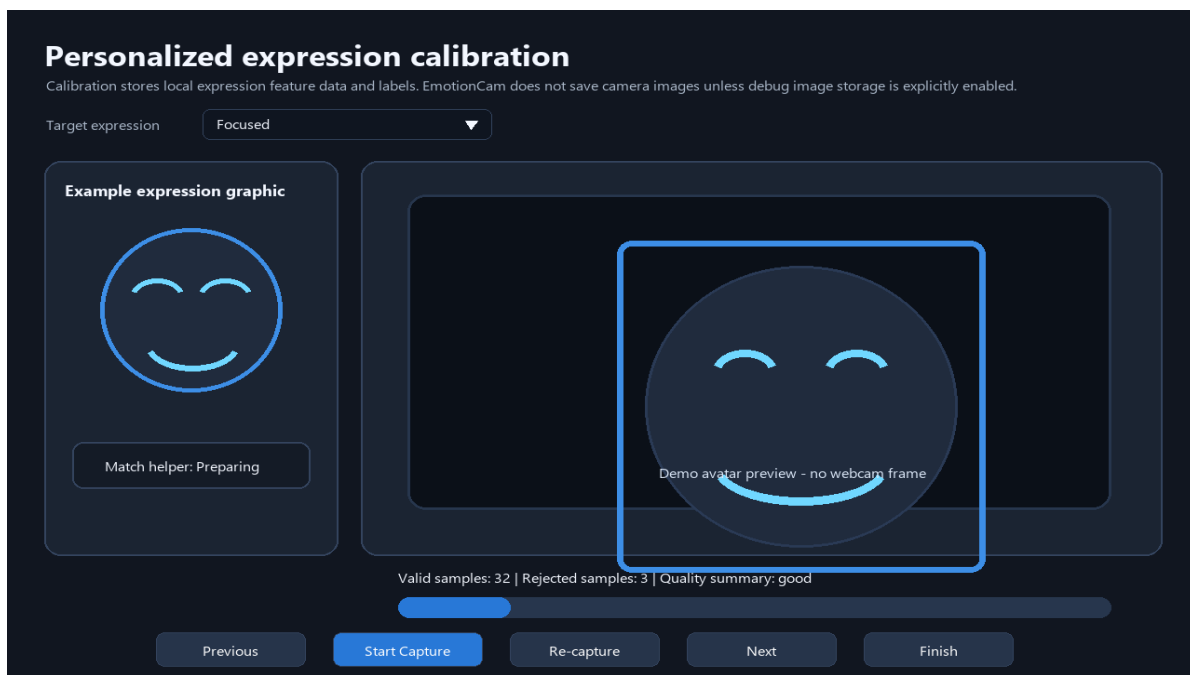
Previous

Start Capture

Re-capture

Next

Finish



10. Settings

Settings include Appearance, User Profile, Daily Email Summary, Camera and Detection, Expression Detection Mode and External AI, Personalized calibration, Background, Privacy/logging, and Debug options.

Use Appearance > Theme to switch between Dark and Light. The theme is saved locally and applies across the dashboard, Settings, calibration, Statistics, buttons, inputs, progress bars, and message areas.

All numeric fields use spinner controls. You can type values or use the visible up/down arrows.

External AI settings include enable/disable, consent, provider, API key, secure-key storage, Test Connection, detection mode, cropped-face-only mode, request interval, timeout, and AI debug info.

EmotionCam Settings

Appearance

Theme

Dark / Light

Interface

Polished dashboard, dialogs, controls

User Profile

User name

Optional

Email address

Optional; used only for enabled summaries

Daily Email Summary

Enabled

Off by default

Delivery

Default mail draft or SMTP

Send-time

20:00

Camera and Detection

Camera index

0

Detection confidence

0.45

Smoothering window

1.5 s

Face-missing grace

1.0 s

SaveCancel

EmotionCam Settings

Appearance

Theme

Dark / Light

Interface

Polished dashboard, dialogs, controls

User Profile

User name

Optional

Email address

Optional; used only for enabled summaries

Daily Email Summary

Enabled

Off by default

Delivery

Default mail draft or SMTP

Send-time

20:00

Camera and Detection

Camera index

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Detection confidence

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1.5 s

Face-missing grace

1.0 s

SaveCancel

EmotionCam Settings

Camera and Detection

Camera index
0

Detection confidence
0.45

Smoothing-window
1.5 s

Appearance

Theme
Dark / Light

Interface
Polished dashboard, dialogs, controls

Save
Cancel

EmotionCam Settings

Appearance

Theme
Dark / Light

Interface
Polished dashboard, dialogs, controls

User Profile

User name
Optional

Email address
Optional; used only for enabled summaries

Daily Email Summary

Enabled
Off by default

Delivery
Default mail draft or SMTP

Send time
20:00

Camera and Detection

Camera index
0

Detection confidence
0.45

Smoothing-window
1.5 s

Face-missing grace
1.0 s

Save
Cancel

11. Statistics

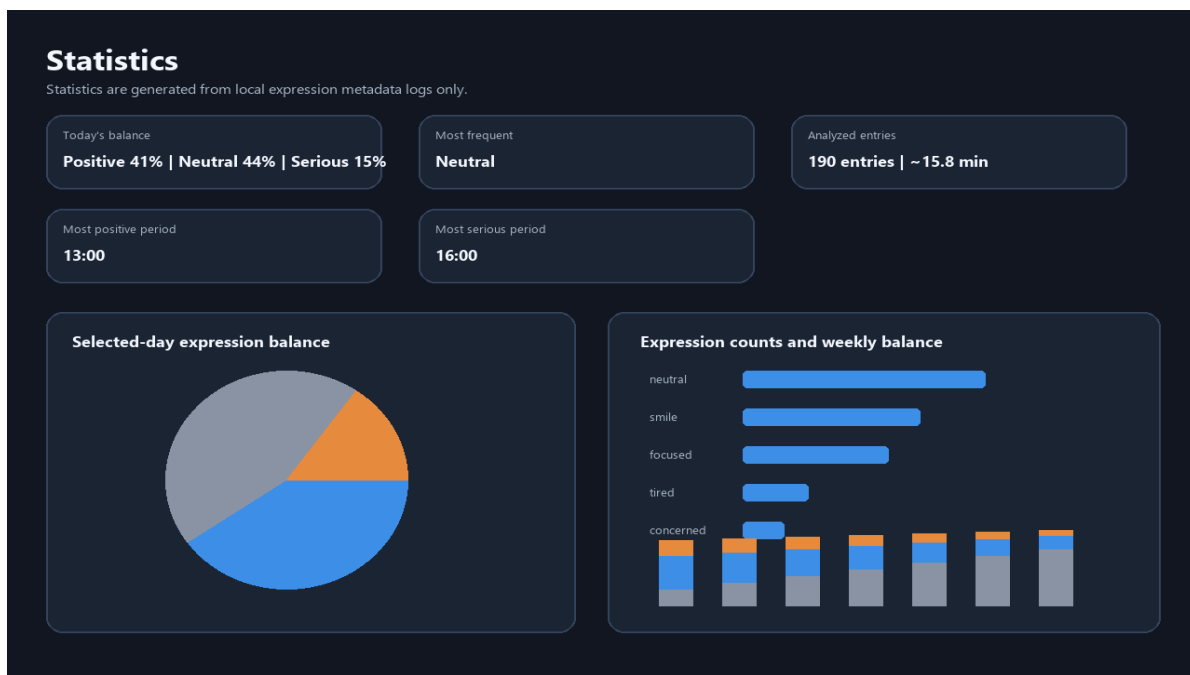
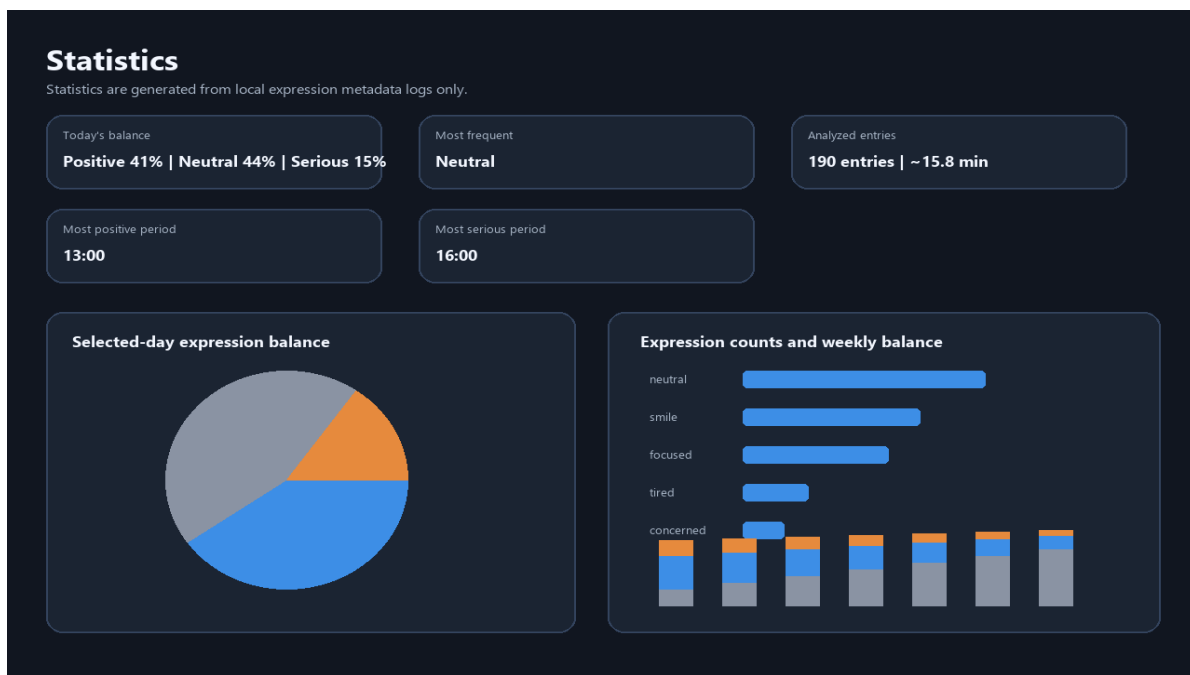
Click Statistics on the dashboard to open local visual summaries generated from expression metadata logs only.

The Statistics window includes:

- Selected-day positive / neutral / serious-expression balance
- Daily expression-group timeline
- Expression counts by label

- Last-7-days balance
- Most frequent expression
- Approximate analyzed entries/minutes
- Most positive and most serious-expression periods when available
- Refresh and Export Summary controls





12. User Profile and Optional Daily Email

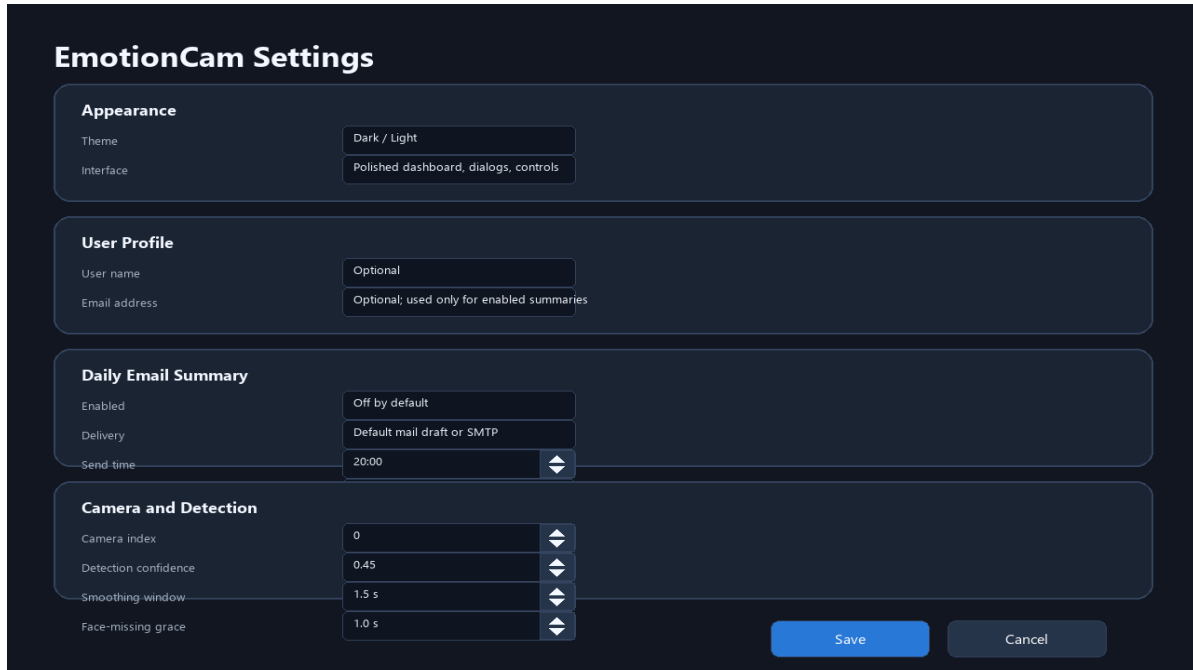
Settings and calibration include optional User name and Email address fields. The name can personalize safe messages. Email is used only for the optional daily summary feature.

Daily email summaries are disabled by default. If enabled, they send local statistics text only. No webcam frames, face images, videos, raw calibration images, or identity data are included.

Delivery modes:

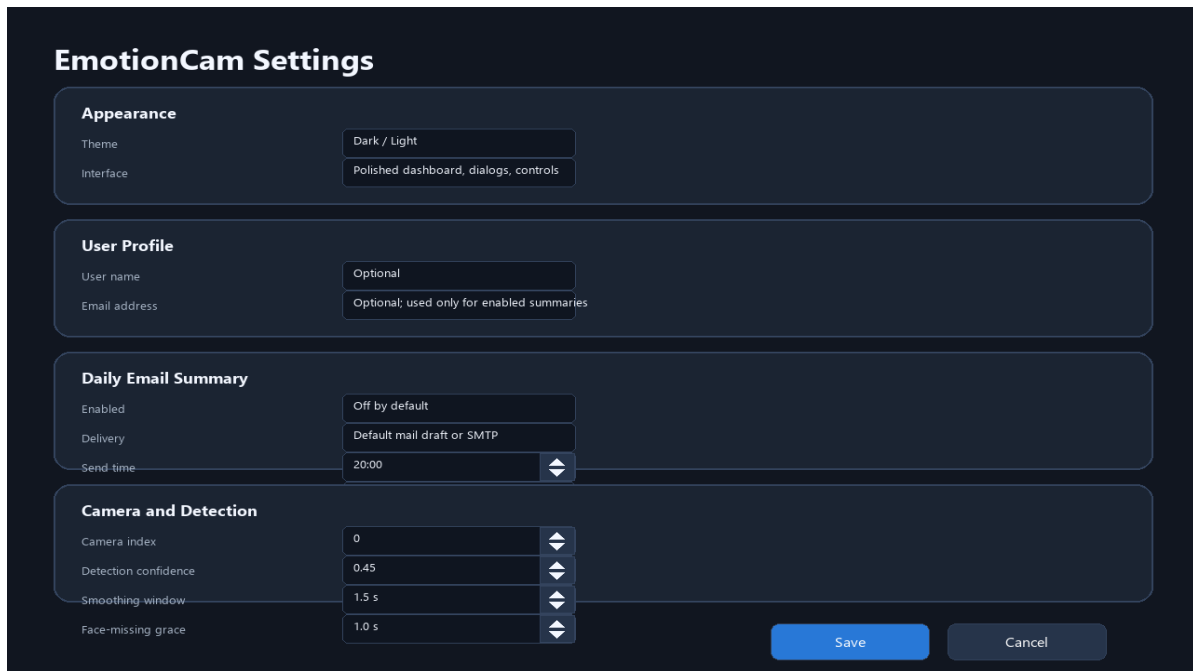
- Default mail client draft: opens a draft for manual review/send.
- SMTP automatic sending: sends at the configured time when the app is running and logs exist.

SMTP passwords are not stored in profile JSON. When Windows credential storage through keyring is available, the password can be stored securely.



The image shows a settings dialog box titled "EmotionCam Settings". It has a dark blue background with white text. The settings are organized into four sections: Appearance, User Profile, Daily Email Summary, and Camera and Detection. Each section contains a list of settings with their current values. At the bottom right, there are "Save" and "Cancel" buttons.

Section	Setting	Value
Appearance	Theme	Dark / Light
	Interface	Polished dashboard, dialogs, controls
User Profile	User name	Optional
	Email address	Optional; used only for enabled summaries
Daily Email Summary	Enabled	Off by default
	Delivery	Default mail draft or SMTP
	Send time	20:00
Camera and Detection	Camera index	0
	Detection confidence	0.45
	Smoothing-window	1.5 s
	Face-missing grace	1.0 s

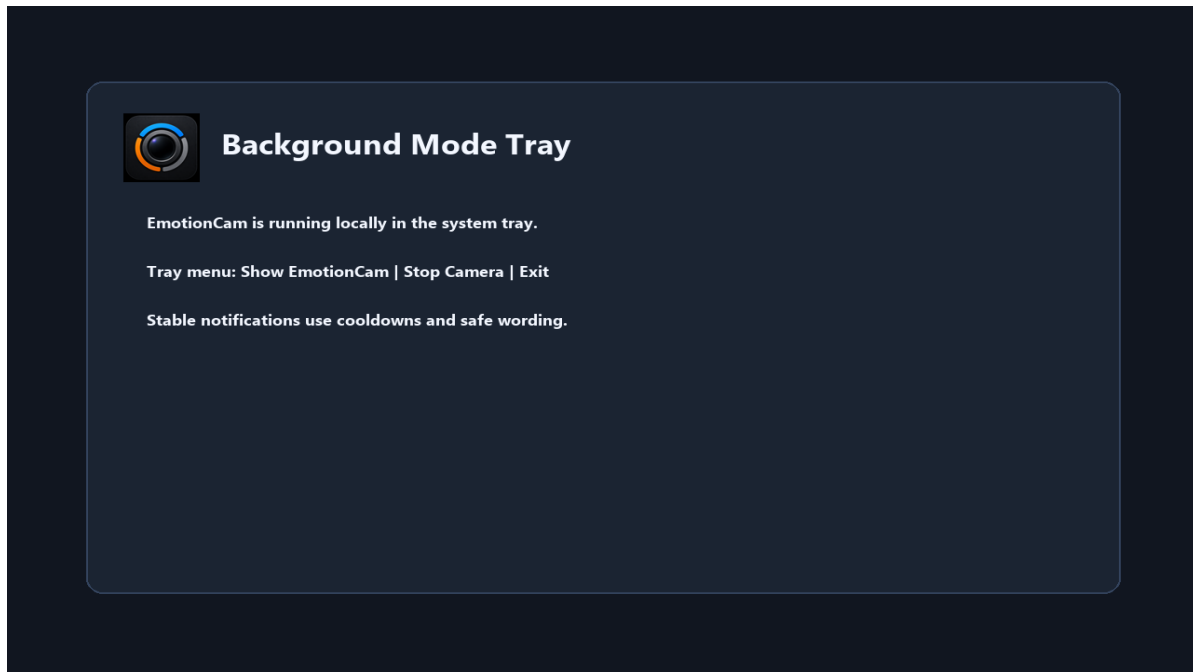


This is a duplicate of the settings dialog box shown above. It contains the same sections and settings: Appearance (Theme: Dark / Light, Interface: Polished dashboard, dialogs, controls), User Profile (User name: Optional, Email address: Optional; used only for enabled summaries), Daily Email Summary (Enabled: Off by default, Delivery: Default mail draft or SMTP, Send time: 20:00), and Camera and Detection (Camera index: 0, Detection confidence: 0.45, Smoothing-window: 1.5 s, Face-missing grace: 1.0 s). The "Save" and "Cancel" buttons are at the bottom right.

13. Background Mode

Click Run in background to keep camera analysis and optional metadata logging active locally while the app hides to the tray when available. The tray menu can show EmotionCam, stop the camera, or exit.

Stable serious-expression estimates can show cooldown-controlled supportive notifications. A later positive shift can show a recovery notification. These are visible-expression estimates, not diagnoses.



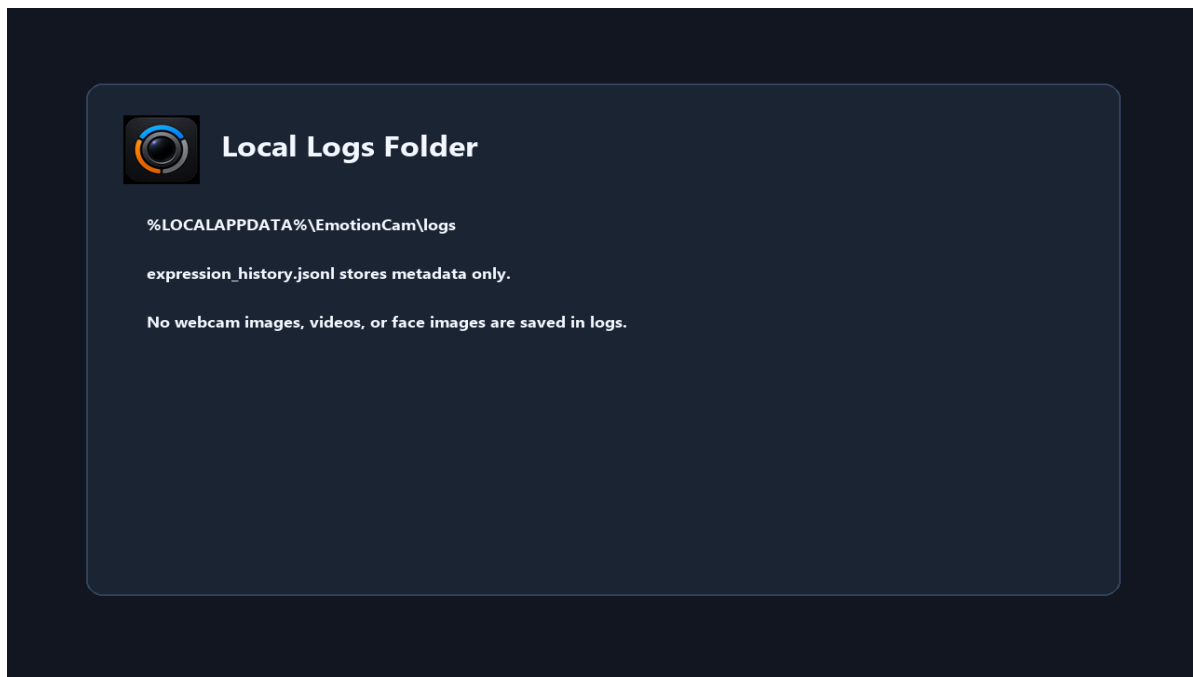
14. Logs and Local Data

Data	Path
Settings	%LOCALAPPDATA%\EmotionCam\config.json
Metadata logs	%LOCALAPPDATA%\EmotionCam\logs\expression_history.jsonl
Personalized profile	%LOCALAPPDATA%\EmotionCam\profile\expression_profile.json
User profile and email preferences	%LOCALAPPDATA%\EmotionCam\profile\user_profile.json
Optional debug calibration images	%LOCALAPPDATA%\EmotionCam\profile\debug_images\

Logs are metadata-only and never include webcam frame data.

AI-related log fields include external_ai_enabled, ai_request_sent, ai_result_label, ai_result_confidence, ai_error, and

final_result_source. Logs never include images, base64 data, or API keys.



15. Demo Walkthrough

- 1 Launch EmotionCam and show the icon/startup privacy screen.
- 1 Start the camera and explain visible-expression estimates.
- 1 Show neutral and positive states; point out the blue positive rectangle.
- 1 Stop the camera and show Camera stopped.
- 1 Open Settings and demonstrate spinner arrows.
- 1 Switch Dark to Light and optionally back to Dark.
- 1 Show Expression Detection Mode and External AI.
- 1 Explain that AI is off by default, requires consent, requires an API key, and uses cropped-face-only mode by default.
- 1 Show Test Connection without entering a real key in public screenshots.
- 1 Switch to Hybrid local + AI and show dashboard AI status/fallback if no key is configured.
- 1 Disable AI again or return to local/hybrid local mode.
- 1 Open calibration and choose an expression from the dropdown.
- 1 Show the local example graphic and match helper.
- 1 Use Start Capture, Re-capture, Previous, Next, and Finish.
- 1 Return to the dashboard and show personalized/hybrid status.
- 1 Run in background and explain tray behavior.

- 1 Open Statistics and show local charts.
- 1 Show User Profile and disabled-by-default email summary settings.
- 1 Open Logs and close the app.

For a presenter-ready version, open START_DEMO_HERE.html.

16. Troubleshooting

Issue	What to try
Camera will not start	Allow desktop-app camera access and close other webcam apps.
No face detected	Improve lighting, face the camera, move closer, and remove occlusion.
Estimate is wrong	Hold steady, reduce head angle, enable Debug, and calibrate.
Calibration has too few samples	Improve lighting, hold still, or increase capture duration.
Background popups do not show	Enable popups, allow Windows notifications, and wait for cooldown.
Statistics are empty	Enable logging, run analysis briefly, refresh, and select a date with entries.
Daily email is not sent	Confirm opt-in, valid email, SMTP settings/password, logs, send time, and network access.
External AI stays off	Check that Enable External AI Analysis and consent are both enabled.
External AI says missing key	Enter an API key or store one securely, then use Test Connection.
External AI errors or times out	Check internet access, API key validity, rate limits, model availability, and timeout setting. Local fallback continues.
Installer is blocked	Use SmartScreen **More info > Run anyway** only for a trusted file.

17. Limitations

- Detection is approximate and may be wrong.
- Lighting, blur, occlusion, distance, and head angle affect results.
- Calibration improves estimates but cannot guarantee accuracy.
- The app is designed for one visible user and one laptop webcam.
- Email summaries require network access only after explicit opt-in.
- External AI analysis requires network access and may send cropped face images or selected frames when explicitly enabled.
- External AI may improve visible-expression estimates but can still be wrong.

18. Uninstall and Data Removal

Uninstall from Windows Settings > Apps > Installed apps > EmotionCam.

To remove remaining local settings, logs, calibration data, and optional debug frames, fully exit EmotionCam and delete:

%LOCALAPPDATA%\EmotionCam